

Ventrapragada Sai Srikar

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Education

University of Maryland College Park

Master of Science in Data Science

Maryland, USA

Aug 2025 – May 2027

Courses: Probability & Statistics, Principles of Data Science and Machine Learning

Mahindra University

Bachelor of Technology in Artificial Intelligence

Hyderabad, India

Aug 2020 – June 2024

Courses: Deep Learning, Natural Language Processing, Reinforcement learning

GPA: 8.52/10

Ranked Top 2% in class of 165 students, Awarded Academic Scholarship for 3 consecutive years

Skills

Programming Languages: Python, C/C++, HTML/CSS, JavaScript, SQL, Bash

Developer Tools: VS Code, Git Version Control, Github, Docker, Vim, Linux

Technologies/Frameworks: Pytorch, JAX, Tensorflow, Keras, Scikit-learn, OpenCV, Langchain, Matplotlib, Seaborn, Pandas, Numpy, SpaCy, NLTK, PySpark, Pycord, FastAPI, mongoDB, MySQL, Langchain, Langflow

ML and DL related: Supervised Learning, Unsupervised Learning, Reinforcement Learning (RL), Deep Learning (DL), Natural Language Processing (NLP), Computer Vision (CV), Generative AI

Experience

Data Analyst Intern

Xometry

Boston, USA

June 2026 – Ongoing

- Designed and built an Agentic AI framework to automate Looker dashboard creation, cutting build time by 45%
- Developed an automated system to analyse competitors across pricing and product differentiation. Translated findings to shape category strategy
- Identified and reduced partner concentration (keyman) risk in 8+ process/material/finish combinations by roughly tripling partner coverage in each

Junior Data Scientist

Astria Digital

Hyderabad, India

Feb 2025 – Jun 2025

- Designed and developed demand and price forecasting ML models for a logistics company and achieved an R^2 score of 0.89 and an MAPE of 24.6%
- Deployed scalable serverless ML pipelines on AWS and automated the process of price forecasting thereby reduced the forecasting time by 70% over a horizon of 10 days.

AI Functional Consultant

Government of Telangana

Hyderabad, India

Sep 2024 – Jan 2025

- Worked in tandem with the IT advisor of the state to build a Generative AI chatbot that streamlined retrieval of federal records and enabled faster access to critical information for policy and decision making.
- Assisted in shaping course curricula for the AI for Grassroots Program, a state led initiative aimed at empowering rural students.

Data Science Intern

Aiden Ai

Hyderabad, India

Jan 2024 – Aug 2024

- Developed AI-powered software leveraging YOLO and QLoRA fine tuned LLMs to automate Unqork application creation, reduced the development time by 67% while incorporating a human in the loop for final refinements.
- Built a proof of concept system to automate meeting minutes by processing Teams recordings, utilizing Whisper for voice transcription and OpenCV for speaker diarization. Integrated it with a Retrieval Augmented Generation(RAG) based chat bot to deliver query responses, post meeting action items and summarized insights.
- Identified inefficiencies in manual server deployment processes, proposed and developed an automated solution using Python and AWS SDK(boto3) to provision and manage EC2 instances.

AI & ML Research Intern

5th Bridge Data Technologies

Hyderabad, India

June 2023 – Aug 2023

- Contributed to the development and deployment of state of the art Software for Payload Based Malicious HTTP Traffic Detection system, by leveraging transfer semi-supervised learning techniques.
- Focused on optimizing precision for the Payload Based Malicious HTTP Traffic Detection system, obtained an accuracy of 70%

Research Projects

Machine Learning based approach for accurate sensing of Dissolved Oxygen for Aquaculture March 2024 – June 2024

- Under the guidance of Dr. Bhargava Rajaram developed a cost-effective, machine learning-based smart sensor using supplementary sensors for accurate dissolved oxygen estimation in aquaculture, reducing costs by approximately 80%.

Self-Organizing Map for Automated Quality Assessment in Algorithmic Music Computation Aug 2023 – Jan 2024

- Under the guidance of Dr. Arya K Bhattacharya and Dr. Prafulla Kalapatapu developed a Self-Organizing Map-based Fitness Function for Evolutionary Algorithms to automate quality assessment in Rasa-aligned music composition, demonstrating high accuracy even with limited data. DOI: 10.1109/ICETC162771.2024.10704148

Projects

OneAIClick.com June 2024

- Developed a no-code GUI solution for fine-tuning open source HuggingFace LLMs, enabling users to finetune models with minimal technical knowledge using parameter-efficient methods. Enabled rapid model deployment and fine-tuning with full data privacy, offering a cost-effective alternative to hiring AI specialists for rapid testing of ideas.

Smart Grid Management using Reinforcement Learning Nov 2023

- Developed and implemented On-Policy Monte Carlo, SARSA, and Q-Learning algorithms to optimize battery storage and power plant production in smart grids, enhancing energy efficiency and reliability.

Optimizing Millimeter Wave Communication with Contextual Bandits Oct 2023

- Designed and implemented contextual bandit algorithms ($\mathcal{E}$ – greedy, UCB and Policy gradient) to optimize beam alignment in millimeter-wave communication, enhancing the reliability and efficiency of vehicular communication systems.

Concurrent Data Management: From Locks to NUMA-Aware Hashmap-Based Solution April 2023

- Engineered a hierarchical Hashmap-based NUMA-aware lock, optimizing memory usage to just one word. Reduced remote cache misses, outperforming NUMA-oblivious approaches by 40% on two-socket machines and 100% on four-socket machines under contention.

Generative Art June 2022

- Created computer-generated art by implementing fundamental mechanical physics concepts such as oscillation and gravitation from the ground up using Processing.

Discord Dining Menu Bot May 2022

- Developed a Discord Dining Bot that notifies dorm residents of meal menu 30 minutes prior mealtimes, streamlined the dining experience and achieved 1500+ daily active users.

Achievements

Academic Scholarship: Awarded merit-based scholarship for being in the top 10% of academic performers in my branch for three consecutive academic years (2020–2023).

Co-authored a paper titled "Self-Organizing Map for Automated Quality Assessment in Algorithmic Music Computation,"

Finalist at T-Works Byte Bending Embedded Challenge: Recognized as a finalist in the national-level T-Works Byte Bending Challenge, an embedded systems hackathon. Secured a position among the top 20 teams out of 600+ teams that participated.

Achieved 1500+ Daily Active Users: Developed a Discord Dining Bot to notify users of dorms meal menus 30 minutes prior mealtimes, streamlining dining experiences

Speaker at AI4Bharat Paper Reading Session 9: Delivered a technical talk on Meta's ImageBind Paper.

Guest Lecture for CS3102 Operating Systems:: Delivered a talk on real-world applications of OS, covering algorithms behind concurrent collaborative systems and real-time scheduling techniques.

Embedded Systems Spy Of Light Challenge: Secured 1st place among 15+ teams

Embedded Systems Escape Room Challenge: Secured 3rd place among 20+ teams

Math Rayleigh Race Challenge: Secured 2nd place among 12+ teams

Positions Of Responsibility

Nvidia

Certified Nvidia Student Ambassador

Mar 2023 – Aug 2024

- Conducted a workshop aimed at harnessing Nvidia SDKs and models to enhance proficiency in Natural Language Processing Tasks
- Coordinated a hackathon in conjunction with Nvidia for the ICETC2023 conference, focusing on the inventive application of Nvidia's extant toolkits for the development and refinement of generative pretrained models.

Enigma: Computer Science Club*Vice Secretary*

Mahindra University

Nov 2021 – Jul 2023

- As a council member of the CS club at my university, spearheaded initiatives such as a session on building algorithms by Dr. Zvi Galil, a Generative Hackathon in collaboration with Nvidia and a Game Jam in collaboration with Ubisoft.

Entrepreneurship and Innovation Club*Co-Founder & Podcast Host of Idea to Impact*

Mahindra University

Aug 2023 – April 2024

- Hosted the initial set of episodes of Idea to Impact, which is a student run podcast by Entrepreneurship and Innovation Club.
- Featured prominent episodes with industry leaders, including the CTO of Zerodha and the CEO of Tworks.

Certificates

Machine Learning Specialization*Supervised Machine Learning, Unsupervised Learning, Recommender Systems and Reinforcement Learning**Aug 2022***AI for Everyone***Fundamental understanding of AI**Apr 2022*